Optoelectronic Lab

Experiment 6: Mode selection in laser diodes: DFB laser

11 Fabry-Perot resonators

The Fabry-Perot resonator is a cavity consisting of two parallel mirrors with the reflectivities r_L , r_R and the transmittivities t_L and t_R at a distance L. The material inside has the refractive index n.

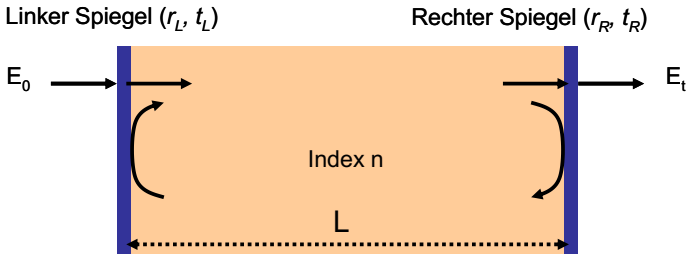


Fig. 1: *Fabry-Perot resonator*

For the following considerations, gains or losses within the resonator (due to stimulated emission or absorption) are neglected.

A light wave, represented by the electric field vector ***E*** with the wavelength λ_0 outside the cavity and λ_0 /n inside, undergoes the phase shift after a complete revolution

$$\phi = 2kL = \frac{4\pi nL}{\lambda_0}; \tag{27}$$

two waves are in phase for relative phase shifts of $\phi = 2m\pi$, m integer.

We are now interested in the electric field E_t of the electric field vector after leaving the cavity (to the right) with a given input of E_0 (from the left). (E_t and E_0 are complex numbers that contain the amplitude and phase information). All reflections and round trips must be taken into account. The result is

$$E_t = E_0 t_L t_R + E_0 t_L r_r r_l t_R e^{i\phi} + E_0 t_L (r_r r_l)^2 t_R e^{i\phi} + \ldots \tag{28}$$

To simplify matters, we set $r_L = r_R$ =r and $t_L = t_R$ =t and introduce the reflection coefficients $R=r^2$ and transmission coefficients $T=t$.[2]

This turns (2) into

$$E_t = E_0 T(1 + R e^{i\phi} + R^2 e^{2i\phi} + \ldots) = \frac{E_0 T}{1 - R e^{i\phi}} \tag{29}$$

(summation geometric series)

This formula already shows very clearly that, in addition to T and R, *the* transmission includes the phase shift, which depends sensitively on L and □!

We obtain the intensity ratio between incident and transmitted radiation by multiplying E_t and E_0 by the respective complex conjugate values, i.e.

$$\frac{I_t}{I_0} = \frac{E_t E_t^*}{E_0 E_0^*} = \frac{T}{(1 - R e^{i\phi})} \cdot \frac{T}{(1 - R e^{-i\phi})} \tag{30}$$

This is a real number, because $e^{i\phi} + e^{-i\phi} = 2\cos\phi$, $T=1-R$ also applies, so

$$\frac{I_t}{I_0} = \frac{T^2}{(1 + R^2 - 2R\cos\phi)} = \frac{(1-R)^2}{(1 + R^2 - 2R\cos\phi)} \tag{31}$$

Usually you reshape a little further and arrive at the *Airy formula* using $(1 - \cos\phi) = 2\sin^2(\phi/2)$

$$\frac{I_t}{I_0} = \frac{(1-R)^2}{(1-R)^2 + 4R\sin^2(\phi/2)} = \frac{1}{1 + \left(\frac{2F}{\pi}\right)^2 \sin^2(\phi/2)}, \tag{32}$$

$F = \frac{\pi\sqrt{R}}{1-R}$ is the so-called finesse of the cavity, which only depends on the reflection coefficient of the mirrors.

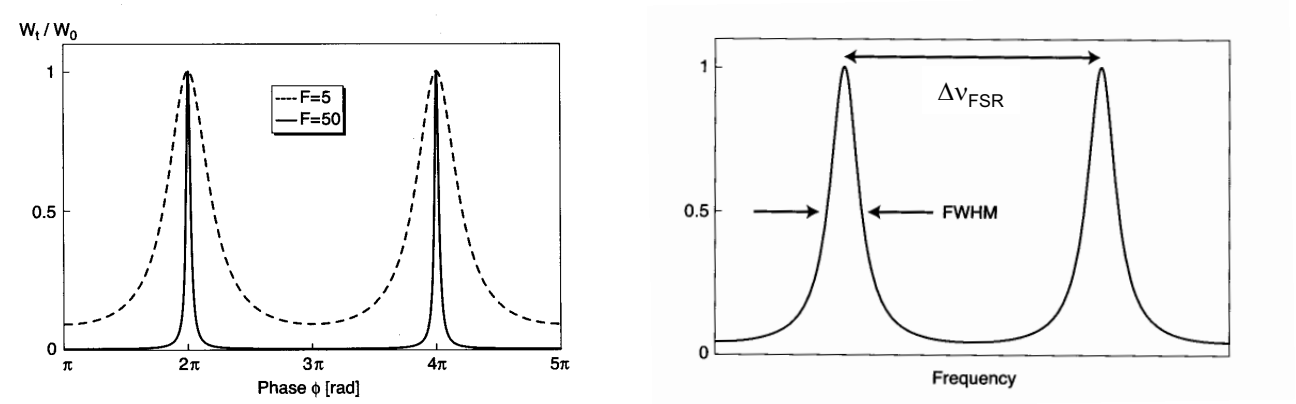


Fig. 2: *Transmission characteristics of a Fabry-Perot resonator.*

Fig. 1 shows that a Fabry-Perot resonator (as expected) exhibits a transmittance of 100% for $\phi = 2m\pi$, m integer, and more or less blocks the transmission in between. The "sharpness" of the transmission peaks is characterized by the finesse. The wavelengths for which the cavity is transmissive result from equation (1) to

$$\lambda_m = \frac{2nL}{m}, \text{ m integer,}$$

the frequencies accordingly

$$\nu_m = \frac{mc}{2nL} \text{ with a distance of } \Delta\nu_{FSR} = \frac{c}{2nL} \tag{33}$$

This value is also referred to as the *free spectral range of* the resonator.

To determine the full width at half maximum (FWHM) of the resonance, we first look for the phase $\phi_{1/2}$, where I/I_{t0} =1/2, which (if we look at the resonance aroundϕ =0, and assume that $\phi_{1/2}$ is small, so that sin( ) $\phi_{1/2} \approx \phi_{1/2}$) is the case for $\phi_{1/2} = \pi / F$ according to equation (6). The FWHM (in "phase units") is now $2\phi_{1/2}$, i.e.

$$FWHM = 2\phi_{1/2} = \frac{2\pi}{F} \text{ on a frequency scale corresponds to}$$

$$FWHM_{\nu} = \frac{\Delta\nu_{FSR}}{F} \tag{34}$$

This equation once again clearly illustrates the finesse as a measure of the half-width.

12 Fashion selection

Conventional semiconductor laser resonators are edge emitters with mirrors at the end of the cavity, i.e. they correspond to the Fabry-Perot resonators described above. In order to achieve a sufficiently high gain, the typical length L of such components is usually in the range of several 100µ m. As the mirrors have no pronounced wavelength dependence, the laser usually oscillates at the energy at which the effective optical gain is maximum or the losses are minimum.

According to (7), the distance between the longitudinal modes is as follows

$$\Delta\nu = \frac{c}{2n_{eff}L} \tag{35}$$

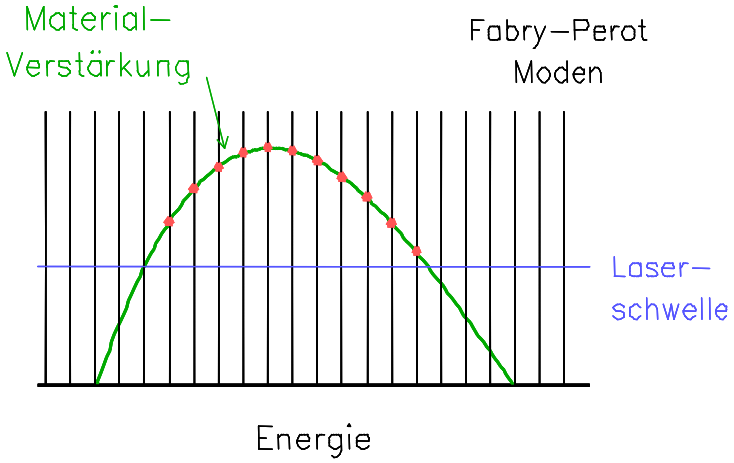


Fig. 3: *Mode spectrum of a Fabry-Perot laser*

In addition, transverse modes must also be taken into account. The relationship between laser geometry and mode spectrum is usually more complicated here and will not be discussed here. Higher transverse modes can be suppressed quite effectively in the laser by using waveguide structures in the laser (see experiment 5) or after the radiation has been emitted by coupling it into special glass fibers. In these structures, higher transverse modes experience strong losses, so that in the ideal case only the so-called TE_{00} mode oscillates.

Longitudinal modes cannot be selected using this method. However, this is possible by using so-called Bragg gratings. This can be used to produce frequency-selective laser diodes or vertically emitting laser diodes with ultra-small threshold currents.

12.1 Principle of frequency-selective laser resonators

The incorporation of frequency-selective elements, such as Bragg gratings, into a laser resonator offers the possibility of producing laser diodes with controlled single-mode emission.

A distinction is made between

- DBR (= Distributed Bragg Reflector) laser diodes
  (Bragg grating is located outside the amplification area)
- DFB (= Distributed Feedback) laser diodes
  (Bragg grating is located within the amplification area)

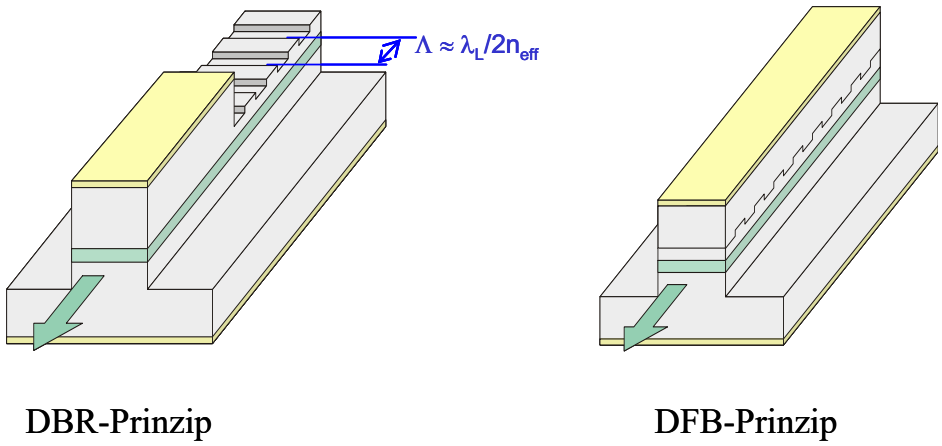


***Fig. 4**: DBR and DFB lasers*

12.2 DBR laser

A *DBR laser* works in exactly the same way as a Fabry-Perot laser. The only difference is that one of the two mirrors of the resonator is replaced by a highly reflective and frequency-selective Bragg grating. This reduces the threshold gain at a selected wavelength, the Bragg wavelength, and the laser oscillates at this wavelength.

A grating with the periodΛ has a characteristic wavelengthλ_{Bragg} at which the reflection has a maximum.

The Bragg condition must be fulfilled for constructive interference of all partial beams reflected on the Bragg grating with the period Λ το οχχυρ:

$$m\lambda_m = 2n_{eff}\Lambda\,,$$

where n_{eff} is the refractive index averaged over the entire layer system. To put it graphically, at this wavelength there is exactly constructive interference of the waves reflected at each sub-stage, so that a maximum of the electromagnetic wave is reflected. The wavelength must therefore "fit" into the grating, so to speak. Compared with equation (7), this is a fairly strict condition: The mode spacing now increases to

$$\Delta\nu = \frac{c}{2n_{eff}\Lambda} \tag{36}$$

This is therefore greater by a factor of Λ/Λ than the free spectral range of the Fabry-Perot resonator with length *L*. The permitted modes are therefore so far apart due to the frequency-selective mirrors that ideally only one mode is active in the laser amplification range.

The Bragg reflector is usually realized with an etched grille.

For flat etched gratings, a small refractive index jumpΔ n is achieved and thus a high frequency selection when using many (typically several thousand) periods.

On the other hand, very few periods are sufficient for laser activity with deeply etched gratings, but the frequency selection is poorer.

12.3 DFB laser

With the DFB laser, the refractive index in the laser-active area itself is modulated with the period Λ. More precisely, the dielectric constantε is modulated, which is related to the complex refractive index $\tilde{n}$ as follows:

$$\tilde{n} = \sqrt{\varepsilon(\omega)} = n + i \cdot \frac{g\lambda_0 c}{4\pi} \tag{37}$$

where *n is* the real refractive index and *g* is the gain in the laser medium (or loss, in which case g is negative).λ_0 is the wavelength of light in a vacuum.

Depending on whether the real or imaginary part of the refractive index $\tilde{n}$ is modulated, a distinction is made between index coupling (Δ n>0) and gain coupling (Δ g>0).

A distinction must therefore be made between the following cases:

- *n* is varied, *g* is constant. The wave therefore undergoes periodic reflections in the laser medium, similar to the reflections on the Bragg grating outside the medium in the DBR laser. This is referred to as *index coupling.*
- *n* is constant, *g* is varied. The wave therefore experiences a locally different amplification or attenuation in the laser medium. Maximum amplification can be expected if the periodicity of *g* also fulfills the Bragg condition. This is referred to as *gain coupling*.
- *n* and *g* are varied, resulting in a combination of index and gain coupling.

Production is usually carried out in 3 technological steps: epitaxy - lithography - epitaxial overgrowth.

In epitaxy, a lattice (e.g. by etching) is introduced above the active layer. The growth process is interrupted for this purpose. Due to the high resolution required, the lattice is usually defined using electron beam lithography. After lateral structuring, the laser is then epitaxially overgrown again.

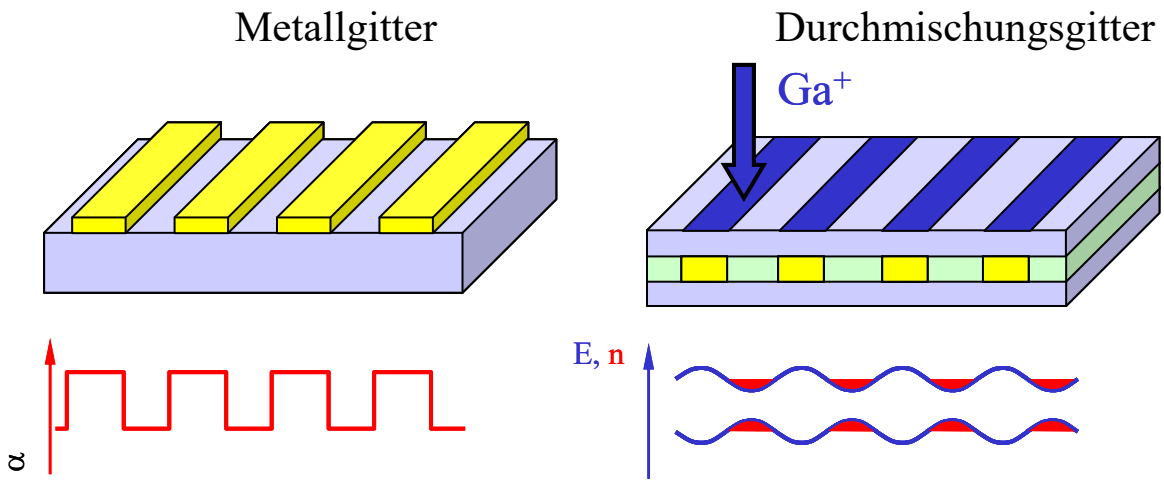


Fig. 5: *Structuring options for DFB lasers*

Other approaches are based on the periodic change of gain or absorption by metal gratings or by implanting ions into the active region.

13 Fine tuning of the wavelength

13.1 Fabry-Perot laser

Fine tuning of the wavelength can be achieved by changing the temperature or changing the current density.

- In semiconductors, an increase in temperature initially causes a reduction in the band gap so that the amplification profile shifts to longer wavelengths. In addition, the resonator expands and the refractive index usually increases so that the modes according to $\lambda_m = \frac{2nL}{m}$ also shift towards the long wavelengths. This effect is dominant, so the mode spectrum shifts below the gain profile. An unpleasant side effect is the so-called mode jumps that can occur.
- A change in the operating current also leads to a shift in the emission wavelength. On the one hand, this is due to the temperature change in the semiconductor caused by ohmic losses; on the other hand, the change in the charge carrier density in the active zone causes a variation in the refractive index and thus also shifts the mode spectrum of the resonator.

13.2 DFB and DBR lasers

The special feature of frequency-selective lasers is that the emission wavelength can be adjusted by changing the grating period. The structuring therefore already allows very fine tuning here.

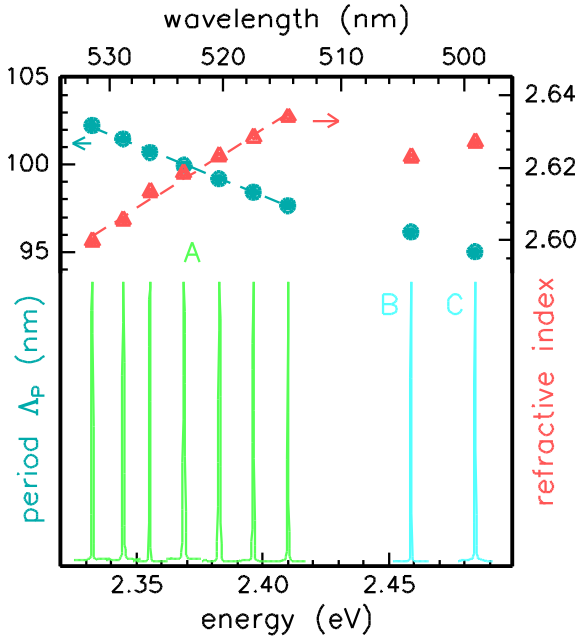


Fig. 6: *Tuning the emission wavelength by changing the grating period using the example of a DBR laser with emission in the green spectral range*

As with the Fabry-Perot laser, the emission wavelength also depends on temperature and charge carrier density. The major difference is that the change in the band edge and also the change in the overall resonator length L are not relevant: instead, the focus is on the temperature and current dependence of the *effective refractive index*. In particular, mode jumps are avoided and the individual mode can therefore be tuned in a controlled and extremely precise manner.

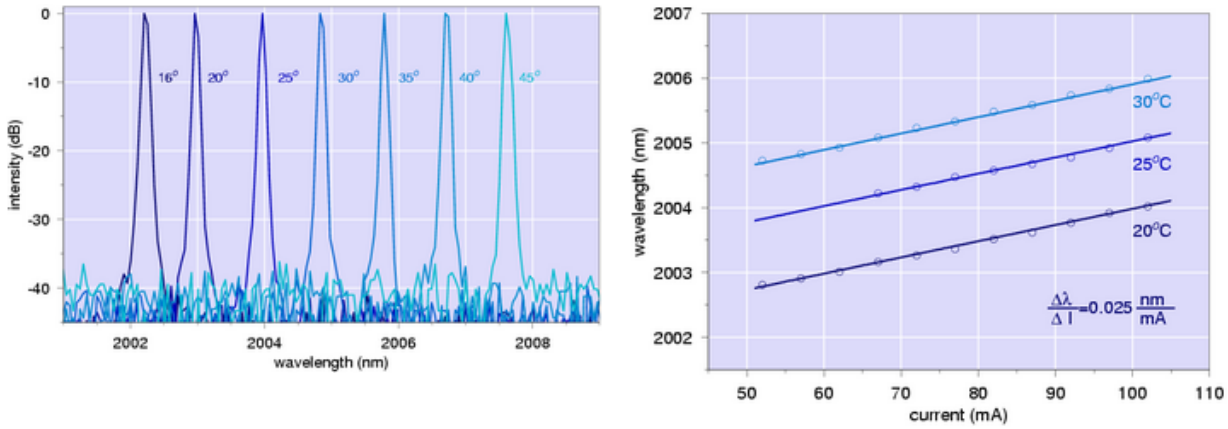


Fig. 7: *Tuning of the emission wavelength by changing the operating temperature (left) and the current (right). Source: www.nanoplus.de*

14 Carrying out the experiment

14.1 Laser diodes

In the practical experiment, a DFB laser diode is to be compared with a laser diode with a Fabry-Perot resonator. The DFB laser emits atλ =1576 nm, the Fabry-Perot laser atλ =1550 nm. The diodes are permanently installed in temperature-controlled holders.

The maximum laser power is 2.5 mW. This radiation can already cause damage to the retina. As laser radiation is invisible, caution is advised! Never look into the beam! Use laser safety goggles!

14.2 Laser diode control

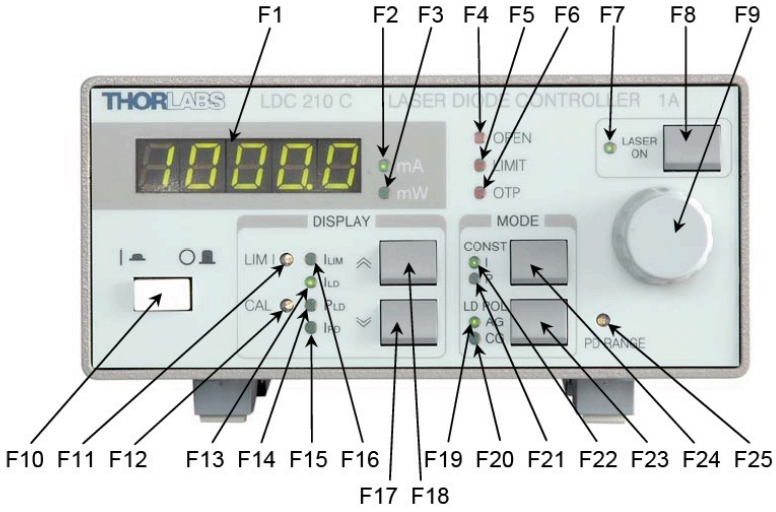


Fig. 8: *Front of the laser diode control*

The diode control is shown in the figure above. Only the constant current mode is used in the experiment:

- Turn controller F9 all the way to the left.
- Select Constant Current Mode by pressing button F24 until diode F22 lights up
- Press button F17 or F18 until diode I_{LD} (F13) lights up
- Switch the diode on via F8
- Set the current via F9. In principle, a current limiter prevents the diode from burning out, but be EXTREMELY careful here. Observe both the display F1 (laser operation is to be expected from approx. 20 mA) and the emission of the laser diode directly in the assembly.
- Press F17 or F18 to display either the current fed in or the laser power decoupled.

14.3 Temperature control

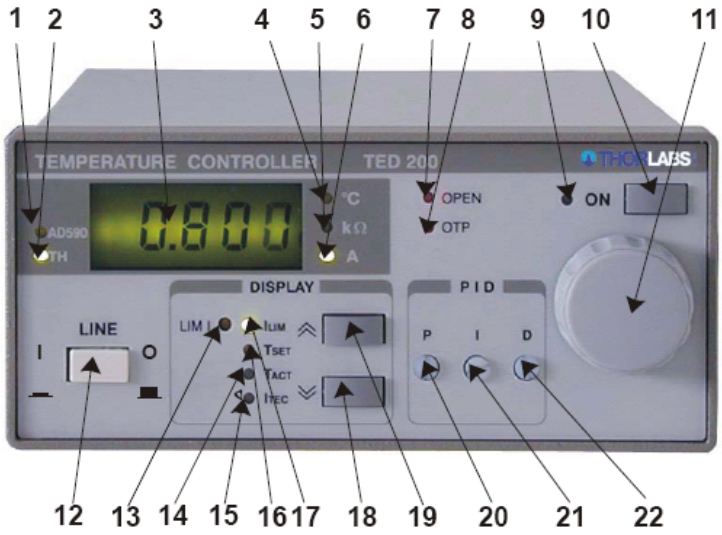


Fig. 9: *Front of the temperature control*

The temperature control shown in Fig. 9 is set to a temperature of approx. 25 °C. To change the temperature:

- Press button 18 or 19 to switch between the display modes I_{LIM} , T_{SET} , T_{ACT} and I_{TEC} (which mode is active is indicated by the respective diode 14-17). Select mode T_{SET}
- Set the temperature using the control dial 11. A small difficulty here: The temperature cannot be set directly, instead you must set the desired resistance of the temperature sensor in kΩ . The temperature values are assigned using the following calibration curve.

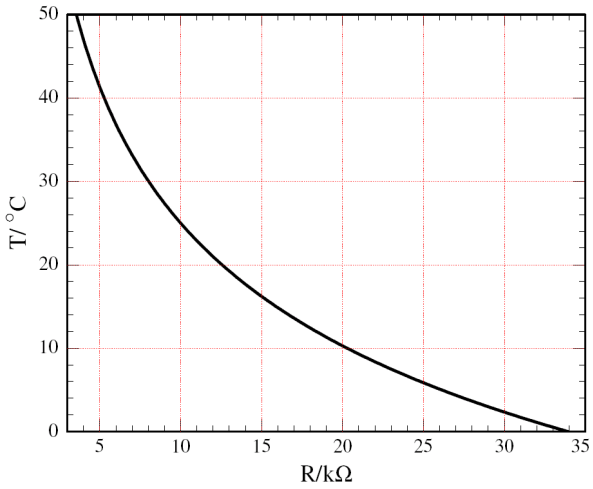


Fig. 10: *Calibration of the temperature control*

- Select the mode T_{ACT} to check the actual resistance and wait until the temperature has stabilized.

14.4 Monochromator and detector

The spectral analysis is carried out with the aid of a grating monochromator.

14.5 Fabry-Perot spectrometer

A so-called Fabry-Perot spectrometer is available for analysis. This is a combination of a Fabry-Perot resonator (see Fig. 1) with a photodiode. The length of the resonator can be set very precisely using a piezoelectric element.

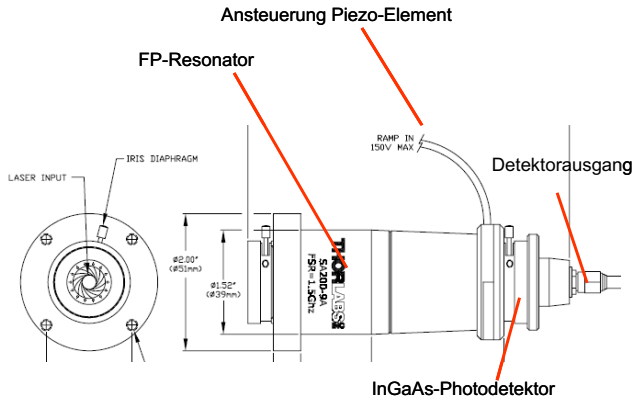


Fig. 11: *Fabry-Perot spectrometer*

If the length L of the resonator fulfills the condition $m\lambda_{Laser} = 2L$ (m integer), the laser light is transmitted and the photodiode detects a signal. If the length is set "incorrectly", however, no signal is detected. The transmission characteristic follows the Airy formula described in section 1 (Fig. 2 and 3). This results in a spectrometer with exceptionally high spectral resolution.

- The resolution of the Fabry-Perot spectrometer used is given as 67 MHz, the free spectral range as 10 GHz. Convert these values into wavelength differences in the wavelength range around 1500 nm. What is the finesse of the spectrometer?

During operation, a sawtooth voltage with an amplitude of around 10 V and a frequency of around 100 Hz is applied to the input for the piezoelectric element. The signal from the photodetector is recorded via an oscilloscope. In this way, the wavelength scale of the laser line is converted into a time scale on the oscilloscope screen.

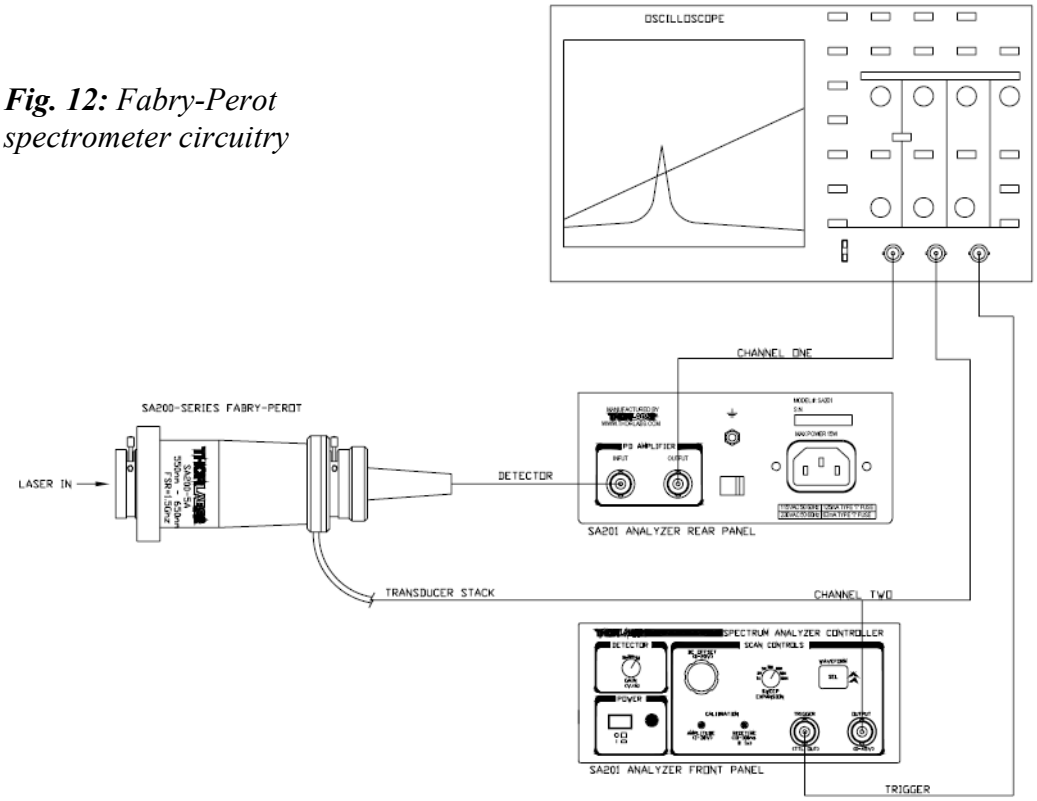


Fig. 12: *Fabry-Perot spectrometer circuitry*

Calibration is simple: Select the amplitude of the sawtooth voltage so that two lines are visible. The distance between the two lines corresponds to the FSR of the spectrometer (**why??**). This can be used to convert the oscilloscope's time axis into wavelengths or frequencies.

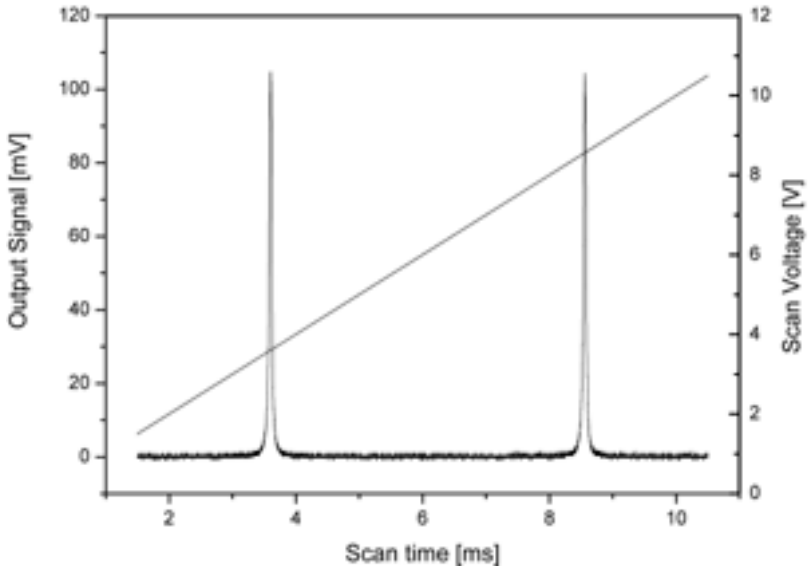


Fig. 13: *Calibration of the oscilloscope time axis via the FSR of the Fabry-Perot spectrometer*

14.6 Tasks

- Connect the DFB laser to the laser diode and temperature controller. Couple the emission of the DFB laser into the Fabry-Perot spectrometer. Make sure that the laser diode is positioned straight in front of the spectrometer. During adjustment, make sure that the lines displayed on the oscilloscope are as narrow as possible (see Fig. 13). Calibrate the time axis of the oscilloscope signal as described above.
- Use the calibration to determine the line width of the DFB laser emission, i.e. the width of the displayed peak at half the maximum intensity (FWHM, Full Width at Half Maximum). Is this above or below the resolution of the spectrometer?
- Now systematically change first the temperature at constant current and then the current at constant temperature of the laser diode. Record at least ten measured values in each case and ensure that you do not exceed the free spectral range of the Fabry-Perot spectrometer. Using the calibration, graph the dependence of the wavelength on the temperature and the current and determine /ΔλΔ T and /ΔλΔ I for the DFB laser.
- Now connect the Fabry-Perot laser to the laser diode and temperature controller and couple the emission of the laser into the Fabry-Perot spectrometer. Describe the differences between the emission spectra of the two lasers. Are mode jumps visible in the Fabry-Perot laser with small changes in current or temperature?
- Analyze the emission of the Fabry-Perot laser. Determine the number of visible modes and the mode spacing.
- What mode spacing do you expect for a resonator length of 1mm (see formula 9, assume n = 3.5)? Can you measure this mode spacing with the Fabry-Perot spectrometer? If not, what kind of signal do you expect? Does this contradict your measurement? If necessary, discuss the experiment with your supervisor.

ATTENTION: (For good reason) The easiest way to convert wavelength differences into frequency differences is to use the formula $\Delta\lambda/\lambda = \Delta\nu/\nu$ (Where does it come from? Hint: error calculation). Never use $\Delta\nu = c/\Delta\lambda$, this formula is WRONG (why?) !!!

When discussing the results in the test protocol, the following should be addressed in particular in each of the points to be measured:

- What results do I expect according to theory and what results do I get in reality?
- How can any deviations between theory and reality be explained (error analysis)?

15 Literature

- Kneubühl, Sigrist, Laser, Teubner (7th edition 2008)
- H. Zappe, Laser Diode Microsystems, Springer Verlag (2004)